

## Lecture 4

### Matrices of Linear Maps; Change of Basis

Throughout, bases are ordered and  $\mathcal{M}(T)$  denotes the matrix of  $T$  with respect to the bases in use. Tags indicate the flavour of each problem; a ★ marks a harder one.

#### Core tutorial route

Work **4.1, 4.7, 4.11, 4.14, 4.15, 4.16, 4.17, and 4.23** in that order. This route covers building the matrix of a map, commuting operators and their matrices, testing invertibility, the dimension of a map space, the equivalence of injectivity, surjectivity and invertibility, computing a matrix in a new basis, converting coordinates, and the proof that trace and determinant are invariant.

**Additional practice:** 4.3–4.4, 4.8–4.10, 4.12–4.13, 4.18–4.22. **Bridge:** 4.2, 4.5, 4.6 (differentiating polynomials). **Extension:** 4.24–4.26.

#### Matrices of maps

**Exercise 4.1** (computational) Let  $T \in \mathcal{L}(\mathbb{R}^3, \mathbb{R}^2)$  be  $T(x, y, z) = (x + 2y - z, 3x + z)$ . Write down the matrix  $\mathcal{M}(T)$  with respect to the standard bases and use it to compute  $[T(1, 1, 1)]$ .

Answer.  $\mathcal{M}(T) = \begin{pmatrix} 1 & 2 & -1 \\ 3 & 0 & 1 \end{pmatrix}$ ;  $[T(1, 1, 1)] = (2, 4)$ .

**Exercise 4.2** (computational) Let  $D \in \mathcal{L}(\mathcal{P}_2(\mathbb{R}))$  be the differentiation operator. In the standard basis  $1, t, t^2$  compute the matrices  $\mathcal{M}(D)$  and  $\mathcal{M}(D)^2$ . Which operator has the matrix  $\mathcal{M}(D)^2$ ?

Answer.  $\mathcal{M}(D) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$ ,  $\mathcal{M}(D)^2 = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \mathcal{M}(D^2)$ .

**Exercise 4.3** (computational) Let  $T \in \mathcal{L}(\mathbb{R}^2)$  be  $T(x, y) = (2x - y, x + 3y)$ . Write down the matrix  $\mathcal{M}(T)$  in the standard basis and use it to compute  $[T(1, 2)]$ .

Answer.  $\mathcal{M}(T) = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}$ ;  $[T(1, 2)] = (0, 7)$ .

**Exercise 4.4** (computational) Let  $T \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^3)$  be  $T(x, y) = (x - y, 2x + y, 4y)$ . Write down the matrix  $\mathcal{M}(T)$  with respect to the standard bases and use it to compute  $[T(3, -1)]$ .

Answer.  $\mathcal{M}(T) = \begin{pmatrix} 1 & -1 \\ 2 & 1 \\ 0 & 4 \end{pmatrix}$ ;  $[T(3, -1)] = (4, 5, -4)$ .

**Exercise 4.5** (computational) Let  $T \in \mathcal{L}(\mathcal{P}_2(\mathbb{R}))$  be  $(Tp)(t) = p(t) + t p'(t)$ . In the standard basis  $1, t, t^2$  write down the matrix  $\mathcal{M}(T)$  and use it to compute  $T(2 - t + 3t^2)$ .

Answer.  $\mathcal{M}(T) = \text{diag}(1, 2, 3)$ ;  $T(2 - t + 3t^2) = 2 - 2t + 9t^2$ .

**Exercise 4.6** (computational) Let  $D \in \mathcal{L}(\mathcal{P}_3(\mathbb{R}))$  be the differentiation operator. Its matrix in the standard basis  $1, t, t^2, t^3$  is

$$\mathcal{M}(D) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Compute  $\mathcal{M}(D)^3$  and find the smallest  $k$  such that  $D^k = 0$ .

**Answer.**  $\mathcal{M}(D)^3 = \begin{pmatrix} 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ ; the smallest  $k$  is  $k = 4$ .

**Exercise 4.7** (proof) Let  $\dim V = n$ , and let  $S, T \in \mathcal{L}(V)$  be operators whose matrices in a fixed basis are  $\mathcal{M}(S), \mathcal{M}(T)$ . Prove that the operators commute,  $ST = TS$ , if and only if their matrices commute,  $\mathcal{M}(S)\mathcal{M}(T) = \mathcal{M}(T)\mathcal{M}(S)$ .

**Hint.** The theorem “A matrix acts on coordinates”, applied twice, gives  $\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T)$ . Then use the fact that the map  $\mathcal{M}$  is injective: operators are equal if and only if their matrices are equal.

**Exercise 4.8** (proof) Fix bases of  $V$  and  $W$ . Prove that for any  $S, T \in \mathcal{L}(V, W)$  and  $a \in \mathbb{F}$ ,  $\mathcal{M}(aS + T) = a\mathcal{M}(S) + \mathcal{M}(T)$ .

**Hint.** Compare the matrices column by column. The  $k$ -th column of  $\mathcal{M}(aS + T)$  is the coordinate vector of  $(aS + T)v_k$ , and the coordinate map is linear.

## Invertibility and isomorphism

**Exercise 4.9** (computational) Decide whether the operator  $T(x, y) = (2x + y, 4x + 2y)$  on  $\mathbb{R}^2$  is invertible. If it is, find  $T^{-1}$ . If not, find a nonzero vector in the kernel  $\ker T$ .

**Answer.** Not invertible ( $\det = 0$ );  $(1, -2) \in \ker T$ .

**Exercise 4.10** (computational) Show that the polynomials  $1, 1 - t, (1 - t)^2$  form a basis of  $\mathcal{P}_2(\mathbb{R})$ . Transport the question to  $\mathbb{R}^3$ .

**Answer.** The coordinate vectors  $(1, 0, 0), (1, -1, 0), (1, -2, 1)$  are linearly independent ( $\det = -1$ ), so the polynomials form a basis.

**Exercise 4.11** (computational) Decide whether the operator  $T(x, y) = (3x + 5y, x + 2y)$  on  $\mathbb{R}^2$  is invertible. If it is, write down a formula for  $T^{-1}$ .

**Answer.** Invertible ( $\det = 1$ );  $T^{-1}(x, y) = (2x - 5y, -x + 3y)$ .

**Exercise 4.12** (computational) Decide whether the operator  $T(x, y, z) = (x + 2y + 3z, 2x + 4y + 6z, x + y + z)$  on  $\mathbb{R}^3$  is invertible. If not, find a nonzero vector in the kernel  $\ker T$ .

**Answer.** Not invertible ( $\det = 0$ );  $(1, -2, 1) \in \ker T$ .

**Exercise 4.13** (computational) Show that the polynomials  $1 + t^2, t + t^3, 1 + t, t^2$  form a basis of  $\mathcal{P}_3(\mathbb{R})$ . Transport the question to  $\mathbb{R}^4$ .

**Answer.** The determinant of the coordinate vectors is  $1 \neq 0$ , so the polynomials are linearly independent and form a basis.

**Exercise 4.14** (proof) Prove that if  $V$  and  $W$  are finite-dimensional, then  $\dim \mathcal{L}(V, W) = (\dim V)(\dim W)$ .

**Hint.** Show that the map  $\mathcal{M}: \mathcal{L}(V, W) \rightarrow \mathbb{F}^{m \times n}$  is an isomorphism (a matrix prescribes exactly the images of the basis vectors), and compute the dimension of the matrix space.

**Exercise 4.15** (proof) Let  $V$  be finite-dimensional and let  $T \in \mathcal{L}(V)$ . Prove that the following are equivalent: (i)  $T$  is injective; (ii)  $T$  is surjective; (iii)  $T$  is invertible.

**Hint.** Use rank-nullity:  $\dim \ker T + \dim \operatorname{im} T = \dim V$ .

### Change of basis and similarity

**Exercise 4.16** (computational) Let  $T(x, y) = (x, -y)$  be the reflection in the  $x$ -axis on  $\mathbb{R}^2$ . Its matrix in the standard basis is  $A = \text{diag}(1, -1)$ . Using the formula  $A' = P^{-1}AP$ , find the matrix of this reflection in the basis  $\mathcal{B}' = ((1, 1), (1, -1))$ .

Answer.  $A' = P^{-1}AP = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ .

**Exercise 4.17** (computational) Let  $\mathcal{B}$  be the standard basis of  $\mathbb{R}^2$  and  $\mathcal{B}' = ((2, 1), (1, 1))$ . The coordinate vector of a vector  $v$  in the new basis is  $[v]_{\mathcal{B}'} = (1, 2)$ . Find  $[v]_{\mathcal{B}}$  (that is, the vector  $v$  itself). Also find the coordinate vector  $[w]_{\mathcal{B}'}$  of the vector  $w = (3, 0)$ .

Answer.  $[v]_{\mathcal{B}} = (4, 3)$ ;  $[w]_{\mathcal{B}'} = (3, -3)$ .

**Exercise 4.18** (computational) The matrix of an operator  $T \in \mathcal{L}(\mathbb{R}^2)$  in the standard basis is  $A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ . Using the formula  $A' = P^{-1}AP$ , find its matrix in the basis  $\mathcal{B}' = ((1, 0), (1, 1))$ .

Answer.  $A' = \text{diag}(2, 3)$ .

**Exercise 4.19** (computational) In  $\mathbb{R}^2$  take the old basis  $\mathcal{B} = ((1, 2), (1, 3))$  and the new basis  $\mathcal{B}' = ((2, 5), (-1, -2))$ . A vector  $v$  has the coordinate vector  $[v]_{\mathcal{B}} = (1, 3)$ , and an operator  $T \in \mathcal{L}(\mathbb{R}^2)$  has the matrix  $\mathcal{M}(T, \mathcal{B}) = \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix}$ . Find the change-of-basis matrix  $P$ , the coordinate vector  $[v]_{\mathcal{B}'}$  and the matrix  $\mathcal{M}(T, \mathcal{B}')$ .

Answer.  $P = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}$ ;  $[v]_{\mathcal{B}'} = (3, 2)$ ;  $\mathcal{M}(T, \mathcal{B}') = \begin{pmatrix} 3 & -2 \\ 2 & -2 \end{pmatrix}$ .

**Exercise 4.20** (computational) In  $\mathbb{R}^3$  take the old basis  $\mathcal{B} = ((1, 0, 1), (0, 1, 1), (1, 1, 1))$  and the new basis  $\mathcal{B}' = ((1, 1, 0), (0, 1, 1), (0, 0, 1))$ . Find the change-of-basis matrix  $P$  and the coordinate vector  $[v]_{\mathcal{B}'}$  of the vector  $v$  with  $[v]_{\mathcal{B}} = (2, -1, 1)$ .

Answer.  $P = \begin{pmatrix} -1 & 0 & 1 \\ -1 & 1 & 1 \\ 2 & 0 & -1 \end{pmatrix}$ ;  $[v]_{\mathcal{B}'} = (3, -3, 5)$ .

**Exercise 4.21** (computational) In  $\mathcal{P}_1(\mathbb{R})$  take the old basis  $\mathcal{B} = (1 - t, 2 - t)$  and the new basis  $\mathcal{B}' = (1 + t, t)$ . An operator  $T \in \mathcal{L}(\mathcal{P}_1(\mathbb{R}))$  has the matrix  $\mathcal{M}(T, \mathcal{B}) = \begin{pmatrix} -1 & -1 \\ 2 & 3 \end{pmatrix}$ . Find  $\mathcal{M}(T, \mathcal{B}')$ .

Answer.  $\mathcal{M}(T, \mathcal{B}') = \begin{pmatrix} 1 & -1 \\ -2 & 1 \end{pmatrix}$ .

**Exercise 4.22** (computational) Using only trace and determinant, decide whether the matrices in each pair can be similar:

$$(a) \quad \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \text{ and } \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}; \quad (b) \quad \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} \text{ and } \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}.$$

Answer. (a) No: the traces differ,  $5 \neq 4$ . (b) The invariants do not rule similarity out: both matrices have trace 4 and determinant 4.

**Exercise 4.23** (proof) Prove that similar matrices have the same determinant and the same trace.

Hint. For the determinant, use  $\det(XY) = \det X \det Y$ . For the trace, use the identity  $\text{tr}(XY) = \text{tr}(YX)$ , grouping the product  $P^{-1}AP$  as  $(P^{-1}A)P$ .

**Exercise 4.24** (★) Suppose the matrix  $A$  is similar to the matrix  $B$ , and  $B$  is similar to  $C$ . Prove that  $A$  is similar to  $C$ . Then prove that similarity is an equivalence relation on the set of  $n \times n$  matrices.

**Hint.** For transitivity, multiply the two matrices that give the similarities and simplify the expression  $(PQ)^{-1}A(PQ)$ .

**Exercise 4.25** (★) Suppose  $A$  and  $B$  are  $n \times n$  matrices and  $A$  is invertible. Prove that the matrices  $AB$  and  $BA$  are similar. Then give an example showing that without the invertibility assumption the matrices  $AB$  and  $BA$  need not be similar.

**Hint.** You need a matrix  $P$  such that  $P^{-1}(AB)P = BA$ . Since  $A$  is invertible, try  $P = A$ . For the example, look for  $2 \times 2$  matrices whose product is the zero matrix in one order but not in the other.

**Exercise 4.26** (★) Give two  $2 \times 2$  matrices whose traces and determinants agree, but which are *not* similar. Prove that they are not similar.

**Hint.** Compare the matrix  $3I$  with a non-diagonal matrix that has the same trace and determinant. What is the product  $P^{-1}(3I)P$ ?

**Answer.** For example,  $\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$  and  $\begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$ : both have trace 6 and determinant 9, but they are not similar.